

# When Does Episodic Memory Help or Hurt LLM Agents? A Diagnostic Study of Transfer Conditions

Anonymous ACL submission

## Abstract

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LLM-based agents are increasingly augmented with episodic memory, yet memory is not a universal enhancement: it helps when past experience transfers, and hurts when similarity retrieves non-transferable advice. We study this through a controlled comparison of ALFWorld (structured household tasks, fixed action grammar) and WebShop (open-ended e-commerce, free-form navigation), plus an SQL diagnostic gate that tests whether syntactic regularity alone is sufficient. Our memory-centric framework stores outcome-validated episodic experiences and retrieves them via proactive *Context Retrieval* (CR) and on-demand hybrid *Tool Retrieval* (TR). On ALFWorld, CR+TR improves over ReAct by up to +19.4pp, approaching Reflexion at one-third the step cost. On WebShop, the *same* system degrades strict success by up to -30.5pp. On SQL, a GPT-5.4 gate shows smaller but consistent degradation: ReAct solves 43/50 tasks, while CR+TR solves 39/50. We attribute these outcomes to three transfer conditions: *task compositionality*, *action space regularity*, and *learning transfer radius*. Episodic memory helps when all three are high; when any are weak, semantically plausible memories can become confident misdirection.

## 1 Introduction

LLM-based agents are increasingly augmented with episodic memory: systems that store past experiences and retrieve them to guide future decisions. Recent work has demonstrated impressive gains from such augmentation (Voyager (Wang et al., 2023), GITM (Zhu et al., 2023), MemoryBank (Zhong et al., 2024), and Generative Agents (Park et al., 2023)) under the implicit assumption that storing and retrieving past experience will improve performance. But does it always?

We argue for a stronger, more conditional thesis: episodic memory improves agents when the

environment supports transfer through high task compositionality, regular action spaces, and broad learning transfer radius. When those conditions do not hold, retrieval can make the agent worse by injecting memories that are semantically similar but operationally wrong.

We investigate this thesis through a controlled comparative study across two interactive environments: ALFWorld (Shridhar et al., 2021), a structured household task environment with compositional goals and a fixed action grammar, and WebShop (Yao et al., 2022), an open-ended e-commerce environment with free-form navigation. We then add SQL as a diagnostic stress test: SQL has a highly regular syntax, but many tasks hinge on brittle semantic choices such as duplicate handling, join inclusivity, output shape, and set semantics. Together, these settings separate *retrieval relevance* from *transfer utility*. On ALFWorld, our memory system improves over the ReAct (Yao et al., 2023) baseline by up to +19.4pp on unseen tasks, nearly matching Reflexion (Shinn et al., 2023) (+21.6pp) at roughly one-third the step cost. On WebShop, the *same* memory system *degrades* corrected strict success by up to 30.5pp. Strikingly, a hard-negative control that retrieves maximally *irrelevant* experiences (-12.5pp) outperforms the properly configured system (-30.5pp), revealing that semantically similar but non-transferable memories are more harmful than obviously irrelevant ones in this setting. This ~47pp swing, consistent across development and held-out splits, is the central empirical finding of this paper. The SQL diagnostic provides additional evidence for the same mechanism: in a trusted GPT-5.4 gate, ReAct solves 43/50 tasks, while CR, TR, CR+TR, and hard-negative CR+TR solve 41/50, 42/50, 39/50, and 41/50 respectively.

Our analysis attributes these outcomes to three environment properties: *task compositionality* (shared sub-goals vs. unique tasks), *action space regularity* (fixed grammar vs. free-form), and *learn-*

ing transfer radius (one learning benefiting many tasks vs. narrow reuse). When all three favor transfer, memory yields substantial gains; when one or more fails, retrieved experiences can actively mislead the agent. SQL sharpens the point: a regular action language does not guarantee transfer if superficially similar memories alter task-specific semantic policies. This points to a practical distinction between retrieval relevance and actionable utility, one that memory-augmented agents must model explicitly. Accordingly, the novelty of this work is primarily empirical and diagnostic: rather than proposing a fundamentally new memory architecture, we use a controlled episodic-memory setup to study when retrieval transfer helps and when it fails.

In summary, our contributions are:

1. A **conditional thesis for episodic memory**: memory helps when task compositionality, action space regularity, and learning transfer radius jointly support reuse; it can hurt when similarity retrieves plausible but non-transferable advice.
2. A **controlled empirical study** showing opposite effects of the same memory system: +19.4pp on ALFWorld vs. -30.5pp strict-success degradation on WebShop, with hard-negative ablations revealing that “relevant” but non-transferable memories can cause more harm than irrelevant ones.
3. **SQL diagnostic evidence** showing that syntactic regularity alone is insufficient: on a trusted GPT-5.4 SQL gate, memory variants underperform ReAct because near-miss memories shift duplicate handling, join inclusivity, output shape, and set semantics.
4. A **diagnostic framework for retrieval transfer**, based on task compositionality, action space regularity, and learning transfer radius, that characterizes when episodic memory is likely to help or hurt LLM agents.

## 2 Related Work

**Tool-augmented and memory-augmented agents.** ReAct (Yao et al., 2023) interleaves reasoning with environment actions, and Reflexion (Shinn et al., 2023) extends this with iterative self-correction. Several systems augment LLM agents with persistent memory: Voyager (Wang

et al., 2023) builds a verified skill library in Minecraft, GITM (Zhu et al., 2023) stores successful goal decompositions, Generative Agents (Park et al., 2023) use a memory stream with reflection, MemoryBank (Zhong et al., 2024) incorporates forgetting curves, A-Mem (Xu et al., 2025) treats memory itself as an agentic module that autonomously decides what to store, update, and retrieve, and EvoSkill (Alzubi et al., 2026) automatically discovers and evolves reusable skills from execution failures. A common thread is the assumption that accumulated knowledge, whether stored as memories or skills, is net-positive. Our work challenges this by showing that the same episodic memory mechanism yields +19pp gains in one environment and -31pp degradation in another.

**Interactive benchmarks.** ALFWorld (Shridhar et al., 2021) provides compositional household tasks with a fixed action grammar and shared sub-goal structure, making past experience highly reusable. WebShop (Yao et al., 2022) simulates e-commerce with free-form search and unique products, where successful strategies rarely transfer. This structural contrast motivates our comparative design.

**Retrieval-augmented generation.** RAG (Lewis et al., 2020) improves outputs by conditioning on retrieved documents, and has been extended to agentic settings with trajectory or demonstration retrieval. While the RAG literature focuses on improving retrieval quality under the assumption that relevant retrieval helps, our results reveal a complementary failure mode: when retrieved experiences are *superficially* similar but *structurally* non-transferable, retrieval actively misleads the agent. To our knowledge, this is the first work to systematically characterize when episodic memory helps versus hurts LLM agent performance, providing a diagnostic framework based on task compositionality, action space regularity, and learning transfer radius.

## 3 Methodology

We evaluate a unified memory-centric framework across two interactive environments with contrasting structural properties, and use SQL as a diagnostic boundary case for regular action syntax.

### 3.1 Environments

**ALFWorld.** ALFWorld (Shridhar et al., 2021) provides a text-based interface to the ALFRED embodied household benchmark. Agents issue actions from a fixed grammar of  $\sim 15$  verb templates (e.g., go to, take, open, heat). Six task types share a compositional goal structure: SEARCH  $\rightarrow$  ACQUIRE  $\rightarrow$  TRANSFORM  $\rightarrow$  PLACE. The small object vocabulary ( $\sim 50$  types) and shared sub-goal phases yield high structural overlap between tasks.

**WebShop.** WebShop (Yao et al., 2022) is an e-commerce navigation environment where agents find and purchase products matching natural-language specifications via `search[query]` and `click[element]` actions. Each task targets a unique product, resulting in minimal structural overlap and an effectively infinite action space.

### 3.2 Base Agent: ReAct

All agents build on ReAct (Yao et al., 2023), interleaving reasoning traces with actions. We use Gemini 2.5 Flash as the base LLM for all experiments.

### 3.3 Memory Systems

Our framework augments ReAct with a persistent knowledge base and two retrieval mechanisms: proactive *Context Retrieval* (CR) at task start and on-demand *Tool Retrieval* (TR) during execution. We emphasize that Context Retrieval (CR) and Tool Retrieval (TR) address complementary aspects of transfer: CR improves state understanding through contextual grounding, while TR enables action-level reuse. Their combination is designed to balance interpretability and execution efficiency.

#### 3.3.1 Knowledge Base Construction

The knowledge base is constructed offline from training trajectories using LLM-based extraction. For each task, we extract *issue-learning pairs*: structured records associating a mistake with its successful correction. Each entry includes task metadata, the issue-learning texts, a validation level reflecting downstream outcome (VALID\_NEXT\_TRIAL, VALID\_SAME\_TRIAL, or CANDIDATE), and provenance information. We assign scalar validation weights  $\omega$  to each level:  $\omega_{\text{next}} = 1.0$ ,  $\omega_{\text{same}} = 0.9$ ,  $\omega_{\text{cand}} = 0.6$ , used by both retrieval mechanisms to prioritize high-confidence learnings. Full schema details are in

Appendix B.

#### 3.3.2 Context Retrieval (CR)

CR operates proactively: before the agent acts, it retrieves relevant past learnings based on embedding similarity (using `gemini-embedding-001`) between the new task description and historical tasks. The top- $k$  ( $k=5$ ) most similar tasks are identified, their associated learnings are pooled and scored by:

$$\text{Score}_{\text{CR}}(\ell) = \text{sim}(\mathbf{v}_q, \mathbf{v}_{t(\ell)}) \times \omega_\ell \quad (1)$$

where `sim` is cosine similarity and  $\omega_\ell$  is the validation weight.

**Dynamic context sizing.** Rather than retrieving a fixed number of learnings, CR adapts the context window to neighborhood complexity. Let  $L_{\text{max}}$  denote the maximum learning count among the top- $k$  tasks. The retrieval quota is:

$$N_{\text{pick}} = \min(\lceil \alpha \cdot L_{\text{max}} \rceil, p) \quad (2)$$

where  $\alpha = 1.5$  is a scaling factor and  $p \in [5, 25]$  is a tunable cap. The top-scored learnings are formatted as issue-learning pairs and prepended to the agent’s prompt. The full procedure is given in Algorithm 1 (Appendix A).

#### 3.3.3 Tool Retrieval (TR)

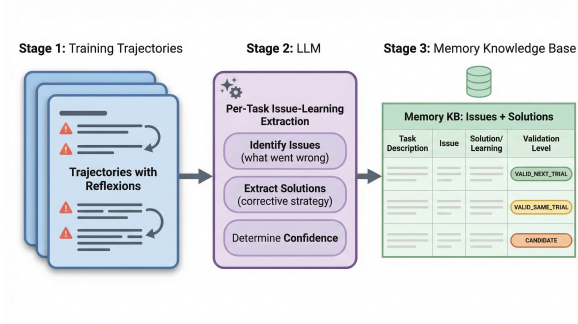
TR provides on-demand retrieval exposed as an action `help["issue"]`. It uses hybrid dense-sparse retrieval: cosine similarity over issue embeddings captures semantic matches, while BM25 (Robertson and Zaragoza, 2009) over concatenated fields captures lexical matches. The candidate pool is the union of the top-100 results from each strategy. Scores are normalized to  $[0, 1]$  and fused:

$$\text{Score}_{\text{TR}}(e) = (0.7 \cdot \text{cos}_{\text{norm}} + 0.3 \cdot \text{BM25}_{\text{norm}}) \times \omega_e \quad (3)$$

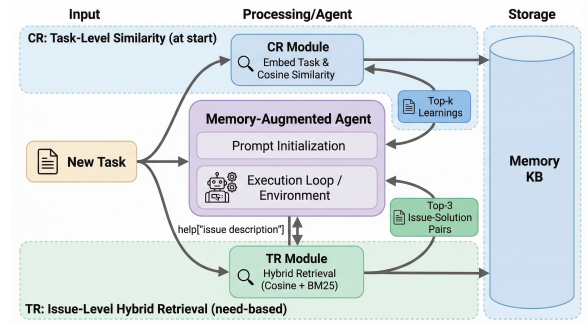
The top- $k$  ( $k=3$ ) results are returned as issue-solution pairs. The full procedure is given in Algorithm 2 (Appendix A).

#### 3.3.4 Hard Negative Control

To verify that gains stem from retrieval *relevance* rather than additional prompt context, we introduce a hard negative control that inverts the selection criterion: CR retrieves the *least* similar tasks and TR selects from the bottom of each retrieval strategy, producing prompts with the same number of learnings but from maximally irrelevant tasks.



(a) Offline KB construction from training trajectories.



(b) Inference-time retrieval via CR and TR.

Figure 1: Overview of the episodic memory framework. (a) Training trajectories with reflexions are processed by an LLM to extract per-task issue–learning pairs with validation levels. (b) Context Retrieval (CR) provides task-level learnings at the start; Tool Retrieval (TR) provides issue-level solutions on demand during execution.

### 3.4 Reflexion Baseline

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We compare against Reflexion (Shinn et al., 2023), an iterative self-correction framework allowing up to 7 trials per task. Unlike our *cross-task* memory (learnings from other tasks), Reflexion is *intra-task* (retrying the same task with accumulated self-critiques), at substantially higher computational cost.

## 4 Experimental Setup

All agent reasoning uses Gemini 2.5 Flash, and all retrieval embeddings use Google’s gemini-embedding-001, held constant across experiments.

**Tasks and splits.** For ALFWorld, the knowledge base is constructed from *alfworld-mini*, a curated subset of 120 training games (20 per task type). We evaluate on the standard validation-seen (140 tasks) and validation-unseen (134 tasks) splits. For WebShop, we evaluate on development (200 tasks) and held-out test (200 tasks) splits, with the knowledge base extracted from separate training tasks. Full dataset details are in Appendix C.

**Framework variants.** We evaluate six variants per environment: ReAct (baseline), ReAct + Reflexion (up to 7 trials), ReAct + CR, ReAct + TR, ReAct + CR + TR, and a hard-negative ablation control. All memory variants execute a single trial per task.

**Metrics.** We report task success, average steps per trial and per task (capturing Reflexion’s cumulative retry cost), and gain from the ReAct baseline in percentage points. For WebShop, success refers to corrected strict success. Full metric definitions

are in Appendix C. All hyperparameters and implementation details are in Appendix C.4 and C.5.

## 5 Results

Tables 1–4 report success, step counts, and gain from the ReAct baseline. The central finding: the same memory system produces substantial gains in ALFWorld and substantial losses in WebShop, while SQL provides diagnostic evidence that regular syntax alone does not make retrieved memories transferable.

### 5.1 ALFWorld: Memory Helps

We ablate the memory system by evaluating CR, TR, and their combination independently.

**Context Retrieval drives large gains.** CR alone raises accuracy by +15.7pp (seen) and +14.2pp (unseen), with consistent gains across splits suggesting generalizable patterns rather than memorized solutions. CR also *reduces* average steps, indicating that proactive guidance makes the agent more efficient.

**Tool Retrieval complements CR, especially on unseen tasks.** TR alone yields a modest +2.1pp on seen tasks but +8.2pp on unseen tasks, suggesting that on-demand issue-level retrieval is most valuable for genuinely novel situations.

**CR+TR approaches Reflexion at one-third the cost.** The combined CR+TR achieves +16.4pp/+19.4pp (seen/unseen), nearly matching Reflexion’s +20.7pp/+21.6pp at roughly one-third the step cost (23 vs. 75–79 steps/task).

**Hard-negative control confirms relevance matters.** Replacing relevant retrievals with maximally irrelevant ones degrades performance to



Table 1: ALFWorld validation-seen results (140 tasks). CR = Context Retrieval, TR = Tool Retrieval. **Bold** = best accuracy.

| Framework                      | Success / Total | Accuracy     | Steps/Trial | Steps/Task | Gain   |
|--------------------------------|-----------------|--------------|-------------|------------|--------|
| ReAct                          | 86 / 140        | 61.43        | 24.29       | 24.29      | —      |
| ReAct + Reflexion <sup>†</sup> | 115 / 140       | <b>82.14</b> | 31.72       | 79.06      | +20.71 |
| ReAct + CR                     | 108 / 140       | 77.14        | 23.09       | 23.09      | +15.71 |
| ReAct + TR                     | 89 / 140        | 63.57        | 26.90       | 26.90      | +2.14  |
| ReAct + CR + TR                | 109 / 140       | 77.86        | 25.41       | 25.41      | +16.43 |
| ReAct + hard-neg CR + TR*      | 78 / 140        | 55.71        | 28.41       | 28.41      | −5.71  |

<sup>†</sup>Max 7 trials; Steps/Task reflects cumulative cost. \*Ablation control retrieving *least*-similar entries.

Table 2: ALFWorld validation-unseen results (134 tasks). CR = Context Retrieval, TR = Tool Retrieval. **Bold** = best accuracy.

| Framework                      | Success / Total | Accuracy     | Steps/Trial | Steps/Task | Gain   |
|--------------------------------|-----------------|--------------|-------------|------------|--------|
| ReAct                          | 83 / 134        | 61.94        | 22.38       | 22.38      | —      |
| ReAct + Reflexion <sup>†</sup> | 112 / 134       | <b>83.58</b> | 30.41       | 74.90      | +21.64 |
| ReAct + CR                     | 102 / 134       | 76.12        | 22.55       | 22.55      | +14.18 |
| ReAct + TR                     | 94 / 134        | 70.15        | 24.93       | 24.93      | +8.21  |
| ReAct + CR + TR                | 109 / 134       | 81.34        | 23.35       | 23.35      | +19.40 |
| ReAct + hard-neg CR + TR*      | 60 / 134        | 44.78        | 33.01       | 33.01      | −17.16 |

<sup>†</sup>Max 7 trials; Steps/Task reflects cumulative cost. \*Ablation control retrieving *least*-similar entries.

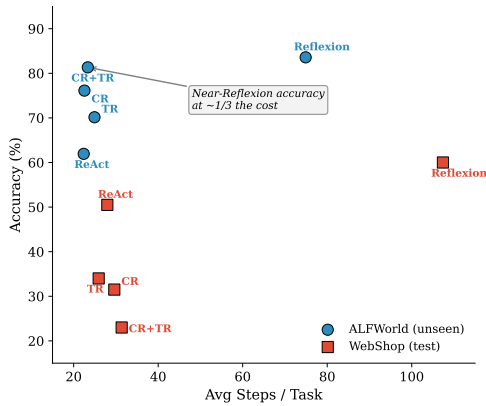


Figure 2: Task success vs. computational cost (steps/task) on held-out splits. CR+TR achieves near-Reflexion success on ALFWorld at  $\sim 1/3$  the cost. On WebShop, memory degrades strict success without reducing cost.

−5.7pp (seen) and −17.2pp (unseen), ruling out prompt-length effects as the source of memory gains.

## 5.2 WebShop: Memory Hurts

**All memory variants degrade strict success.** CR reduces strict success by −29.0pp (dev), TR by −21.5pp, and CR+TR by −30.5pp. Adding

more retrieved context produces monotonically worse outcomes, suggesting retrieved learnings introduce spurious heuristics from structurally dissimilar tasks.

**The hard-negative paradox.** Strikingly, the hard-negative variant (−12.5pp dev, −15.0pp test) *outperforms* all properly configured retrieval variants. We hypothesize that obviously irrelevant retrievals are easier for the agent to recognize and ignore, whereas semantically similar but non-transferable experiences are treated as trustworthy guidance, producing worse outcomes through confident misdirection.

**Only Reflexion improves, at high cost.** Reflexion achieves +11.0pp (dev) and +9.5pp (test) as the only variant surpassing baseline, but at 3.5–3.8 $\times$  the step cost. Its intra-task nature circumvents the transfer problem entirely.

**Test-set confirmation.** All findings replicate on held-out test splits (Table 4), confirming that the observed patterns are robust.

## 5.3 Cross-Environment Comparison

The same memory system, base LLM, and knowledge base pipeline produce a  $\sim 47$ pp swing in ef-

Table 3: WebShop development set results (200 tasks) under corrected strict-success scoring. CR = Context Retrieval, TR = Tool Retrieval. **Bold** = best strict success.

| Framework                      | Success / Total | Strict Success (%) | Steps/Trial | Steps/Task | Gain   |
|--------------------------------|-----------------|--------------------|-------------|------------|--------|
| ReAct                          | 112 / 200       | 56.00              | 24.44       | 24.44      | —      |
| ReAct + Reflexion <sup>†</sup> | 134 / 200       | <b>67.00</b>       | 24.64       | 85.13      | +11.00 |
| ReAct + CR                     | 54 / 200        | 27.00              | 29.39       | 29.39      | −29.00 |
| ReAct + TR                     | 69 / 200        | 34.50              | 23.51       | 23.51      | −21.50 |
| ReAct + CR + TR                | 51 / 200        | 25.50              | 30.52       | 30.52      | −30.50 |
| ReAct + hard-neg CR + TR*      | 87 / 200        | 43.50              | 28.66       | 28.66      | −12.50 |

<sup>†</sup>Max 7 trials; Steps/Task reflects cumulative cost across all trials. \*Ablation control retrieving *least*-similar entries.

Table 4: WebShop test set results (200 tasks) under corrected strict-success scoring. CR = Context Retrieval, TR = Tool Retrieval. **Bold** = best strict success.

| Framework                      | Success / Total | Strict Success (%) | Steps/Trial | Steps/Task | Gain   |
|--------------------------------|-----------------|--------------------|-------------|------------|--------|
| ReAct                          | 101 / 200       | 50.50              | 27.97       | 27.97      | —      |
| ReAct + Reflexion <sup>†</sup> | 120 / 200       | <b>60.00</b>       | 26.81       | 107.38     | +9.50  |
| ReAct + CR                     | 63 / 200        | 31.50              | 29.62       | 29.62      | −19.00 |
| ReAct + TR                     | 68 / 200        | 34.00              | 25.92       | 25.92      | −16.50 |
| ReAct + CR + TR                | 46 / 200        | 23.00              | 31.36       | 31.36      | −27.50 |
| ReAct + hard-neg CR + TR*      | 71 / 200        | 35.50              | 31.43       | 31.43      | −15.00 |

<sup>†</sup>Max 7 trials; Steps/Task reflects cumulative cost across all trials. \*Ablation control retrieving *least*-similar entries.

fect size: CR+TR yields +19.4pp on ALFWorld unseen versus −27.5pp on WebShop test. This divergence is consistent across both split pairings (46.9pp gap in each), confirming it is attributable to environment properties rather than evaluation-set idiosyncrasies.

Figure 3 visualizes this divergence directly: on ALFWorld, all memory variants (CR, TR, CR+TR) exceed the ReAct baseline (dashed line), with CR+TR approaching Reflexion’s success rate. On WebShop, the pattern inverts: every memory variant falls below the baseline, while Reflexion remains the sole method that improves over ReAct. The mirrored structure across paired splits (seen/dev and unseen/test) confirms that the effect is systematic rather than split-specific.

Figure 4 quantifies these effects as gain from the ReAct baseline across all four evaluation splits. The contrast is stark: ALFWorld variants cluster in the positive region (green/blue), while WebShop variants cluster in the negative region (orange/red). Notably, the hard-negative control on WebShop (−12.5/−15.0pp) sits *above* the properly configured retrieval variants (−30.5/−27.5pp), visually highlighting the paradox that relevant retrieval causes more harm than irrelevant retrieval

in low-transfer environments.

The hard-negative controls reveal an asymmetry: on ALFWorld, inverting retrieval degrades accuracy by 22–37pp relative to correct CR+TR; on WebShop, the hard-negative *outperforms* correct retrieval by 12–18pp under strict-success scoring. Reflexion succeeds in both environments (+21pp ALFWorld, +10pp WebShop) because intra-task self-correction does not depend on cross-task transfer, but at 3–4× the step cost.

#### 5.4 SQL Diagnostic Gate

The SQL diagnostic is not used as a third headline benchmark; it is a targeted test of the framework’s middle condition, action space regularity. SQL is highly regular at the syntactic level, yet memory still underperforms ReAct on the trusted GPT-5.4 gate. ReAct solves 43/50 tasks with 0.8999 average reward. CR solves 41/50, TR solves 42/50, CR+TR solves 39/50, and hard-negative CR+TR solves 41/50.

This pattern matters because it rules out a simple explanation in which memory succeeds whenever the action language is structured. In this gate, all memory variants make fewer average steps than ReAct, but the shorter trajectories are not better

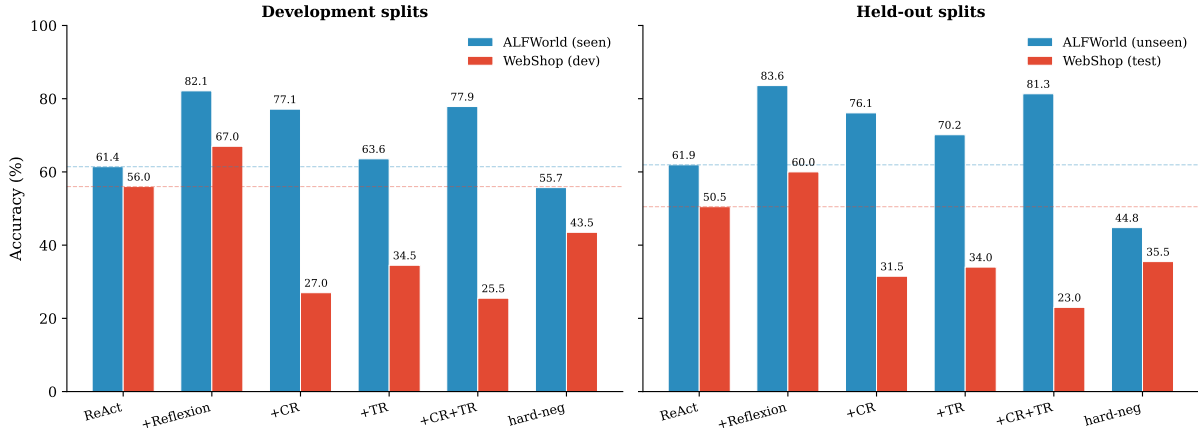


Figure 3: Task success across framework variants with paired splits: development (ALFWorld seen vs. WebShop dev) and held-out (ALFWorld unseen vs. WebShop test). Dashed lines indicate ReAct baselines.

Table 5: SQL transfer-radius diagnostic on the trusted GPT-5.4 gate (50 tasks). Success is exact task success; reward is the environment reward. Help calls are retrieved-tool calls, which are zero in this gate, so TR primarily measures a prompt/tool-availability perturbation rather than useful on-demand retrieval.

| Framework                | Success / Total | Reward | Avg. Steps | Regressions vs. ReAct | Improvements vs. ReAct | Help Calls |
|--------------------------|-----------------|--------|------------|-----------------------|------------------------|------------|
| ReAct                    | 43 / 50         | 0.8999 | 7.48       | —                     | —                      | —          |
| ReAct + CR               | 41 / 50         | 0.8688 | 5.56       | 3                     | 1                      | 0          |
| ReAct + TR               | 42 / 50         | 0.8911 | 5.82       | 1                     | 0                      | 0          |
| ReAct + CR + TR          | 39 / 50         | 0.8361 | 5.72       | 4                     | 0                      | 0          |
| ReAct + hard-neg CR + TR | 41 / 50         | 0.8811 | 5.94       | 2                     | 0                      | 0          |

trajectories: near-miss memories shift latent query policies and reduce exact task success. The result supports the broader thesis that action regularity must be paired with compositional tasks and broad transfer radius.

## 6 Analysis and Discussion

We trace the divergence between environments to three structural properties and synthesize them into a diagnostic framework. SQL then probes an important boundary case: high syntactic regularity with brittle semantic transfer.

### 6.1 Task Compositionality

ALFWorld tasks decompose into shared sub-goal phases (SEARCH → ACQUIRE → TRANSFORM → PLACE) that recur across task types, so a modest set of phase-specific learnings covers a combinatorially large task space. WebShop tasks are monolithic: each targets a unique product through a unique catalog path with no shared sub-task decomposition. This explains why CR yields +15.7pp on ALFWorld but −29pp on WebShop, and why the hard-negative paradox emerges: in WebShop, obviously irrelevant retrievals cause less harm than plausible-looking but non-transferable ones.

**Principle.** Episodic memory is most effective when tasks share compositional sub-task structure.

### 6.2 Action Space Regularity

ALFWorld’s fixed grammar of ~15 verb templates ensures that learned action sequences generalize across tasks. WebShop’s action space is syntactically simple but combinatorially vast: free-form search queries and dynamically generated page elements mean that specific navigation paths never recur. TR’s issue-level guidance accordingly helps in ALFWorld (+8.2pp unseen) but hurts in WebShop (−21.5pp).

**Principle.** Episodic memory is most effective when the action space is constrained and regular.

### 6.3 Learning Transfer Radius

We define the *learning transfer radius* as the number of distinct future tasks that benefit from a single learned experience. In ALFWorld, a learning like “open the container before taking an object” applies to dozens of tasks across multiple types (high radius). In WebShop, each learning is tied to a unique product (radius near zero). The empirical signature is visible in ALFWorld’s stronger

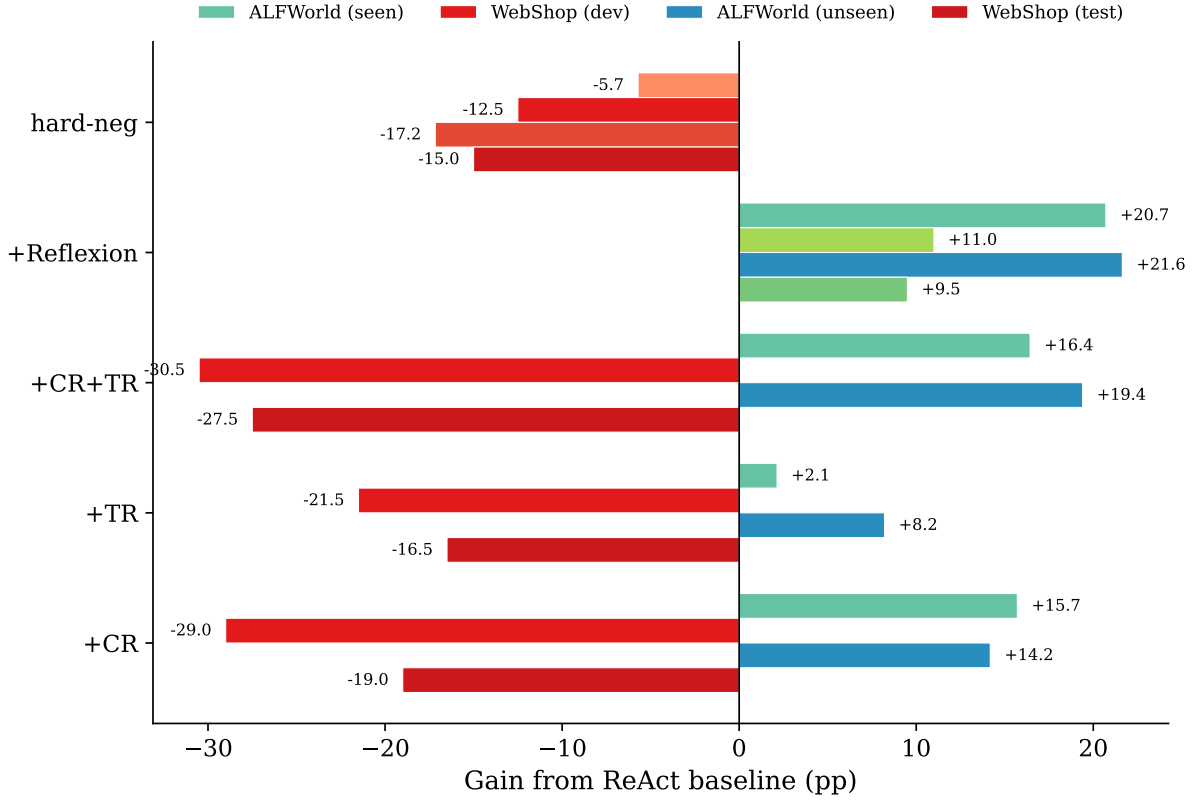


Figure 4: Gain from ReAct baseline (percentage points) for each memory variant across all four evaluation splits. Green/blue indicates improvement; orange/red indicates degradation. Development splits (ALFWorld seen, WebShop dev) and held-out splits (ALFWorld unseen, WebShop test) show consistent patterns, confirming that the  $\sim 47$ pp swing between environments is robust.

unseen-split gains (+19.4pp vs. +16.4pp on seen), indicating effective cross-task generalization.

**Principle.** Episodic memory is most effective when the learning transfer radius is high.

#### 6.4 SQL: Regular Syntax, Brittle Semantics

SQL initially appears favorable for episodic memory because its action space is constrained by a formal grammar. The diagnostic gate shows why this is insufficient. SQL memories often transfer at the level of syntax while failing at the level of denotation: a plausible prior learning can change whether duplicate rows are preserved, whether a join should include zero-count entities, whether exclusion applies to rows or grouped entities, which output columns the grader expects, or whether literal date strings should be normalized.

The strongest regressions are near-miss transfers. In one count query, memories about empty tables and count verification push the agent toward a left join, adding zero-concert stadiums that the reference query excludes. In another, a memory

about countries that do not speak English transfers to TV-channel rows and collapses valid duplicate (aspect ratio, country) rows with DISTINCT. A third case adds a plausible explanatory note column even though the task asks only for death and injury values. These are not syntax failures; they are semantic policy shifts with large reward consequences.

The SQL hard-negative control performs closer to ReAct than CR+TR (41/50 vs. 39/50), echoing the WebShop pattern at smaller scale. Irrelevant memories are often easier to ignore than adjacent memories that look applicable but carry the wrong latent policy. SQL therefore strengthens the framework rather than weakening it: action space regularity is necessary but not sufficient. Memory helps when regular actions are embedded in compositional tasks whose learnings have broad transfer radius.

**Principle.** Syntactic regularity does not guarantee memory transfer; the reusable unit must also preserve the task’s semantic policy.



Table 6: Diagnostic framework for episodic memory effectiveness. Effect sizes are relative to ReAct baseline (seen/unseen for ALFWorld, dev/test for WebShop, trusted GPT-5.4 gate for SQL).

| Property                 | ALFWorld                           | WebShop               | SQL                      |
|--------------------------|------------------------------------|-----------------------|--------------------------|
| Task Compositionality    | High (shared sub-goals)            | Low (unique products) | Medium (query motifs)    |
| Action Space Regularity  | High (fixed, $\sim 15$ verbs)      | Low (free-form)       | High (formal grammar)    |
| Learning Transfer Radius | High ( $1 \rightarrow$ many tasks) | Near-zero             | Narrow (semantic policy) |
| Memory Effect (CR+TR)    | +16.4/+19.4pp                      | -30.5/-27.5pp         | -4/50                    |
| Hard-neg Effect*         | -5.7/-17.2pp                       | -12.5/-15.0pp         | -2/50                    |

\*Hard-neg *outperforms* proper retrieval on WebShop and SQL, indicating semantic similarity can be counterproductive when transfer radius is narrow.

## 6.5 Efficiency–Accuracy Trade-off

Figure 2 plots task success against computational cost (steps per task) on held-out splits, revealing a key practical distinction. On ALFWorld, CR+TR occupies an efficient frontier: it achieves near-Reflexion accuracy (+19.4pp vs. +21.6pp) at roughly one-third the step cost ( $\sim 23$  vs.  $\sim 79$  steps/task), making it the preferred choice when compute budgets are constrained. On WebShop, memory variants shift the agent into a strictly dominated region: strict success degrades without any compensating reduction in cost. Reflexion improves success in both environments but consistently incurs 3–4 $\times$  the baseline step cost.

## 6.6 Diagnostic Framework

Table 6 summarizes how the evaluated settings differ along these dimensions.

When all three properties favor transfer, memory provides substantial gains; when they do not, retrieved experiences can actively mislead the agent. Before deploying a memory system, practitioners should assess their target environment along these dimensions: task compositionality, action space regularity, and learning transfer radius. When these properties are unfavorable or mixed, alternatives such as intra-task self-correction or memory systems with explicit transfer gating may be more appropriate.

## 7 Limitations

The framework is intentionally diagnostic, and it needs more environments before it should be treated as a general predictor. ALFWorld and WebShop provide the main controlled comparison, while SQL is additional evidence for a specific boundary case: regular action syntax with narrow semantic transfer radius. This is stronger than a two-point anecdote, but it is not yet a full map of the space.

We also evaluate a limited set of base-model conditions, and most memory variants use single-trajectory evaluation without repeated-run confidence estimates. Different LLMs may vary in how strongly they follow retrieved memories, how well they discount irrelevant context, and how often they recover from near-miss advice. These model interactions are part of the phenomenon rather than a nuisance variable, and future work should measure them directly.

The next high-value test is ScienceWorld (Wang et al., 2022). It is an orthogonal environment to ALFWorld and WebShop: scientific tasks should provide compositional procedures and reusable causal knowledge, but with different objects, dynamics, and failure modes from household manipulation. We have not run or reported ScienceWorld results here. Its role is future validation of the proposed transfer conditions, especially whether broad learning transfer radius predicts memory benefit outside ALFWorld.

## 8 Conclusion

This work provides evidence that episodic memory is not a universal enhancement for LLM-based agents. Through a controlled study applying an identical memory system to ALFWorld and WebShop, we observe that the same retrieval mechanisms produce a  $\sim 47$ pp swing in effect: up to +19.4pp improvement versus -30.5pp degradation. SQL adds a focused diagnostic: even in a regular action language, CR+TR falls from 43/50 to 39/50 when near-miss memories alter brittle semantic policies. Across these settings, irrelevant memories can cause *less* harm than semantically “relevant” ones, suggesting confident misdirection rather than noise injection as the failure mode.

We attribute this divergence to three structural properties: task compositionality, action space regularity, and learning transfer radius. The practical thesis is direct: memory helps when all three support transfer; when they do not, retrieval should be gated, filtered, or replaced by intra-task correction. Future directions include adaptive memory gating that suppresses retrieval when transfer confidence is low, validation on additional domains and models, ScienceWorld as a planned high-transfer orthogonal test, and memory filtering strategies that assess transferability before injecting retrieved content.

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## Appendix

### A Retrieval Algorithms

**Algorithm 1** Context Retrieval from Knowledge Base

---

**Require:** Task description  $q$ ; knowledge base  $\mathcal{K}$ ; neighbor count  $k=5$ ; scaling factor  $\alpha=1.5$ ; cap  $p \in [5, 25]$ ; validation weights  $W$

**Ensure:** Selected learnings  $C$

- 1:  $\mathbf{v}_q \leftarrow \text{EMBED}(q)$
- 2:  $\mathcal{T}_{\text{top}} \leftarrow \text{Top-}k \text{ tasks in } \mathcal{K} \text{ by } \text{sim}(\mathbf{v}_q, \mathbf{v}_t) \text{ \{filter exact matches\}}$
- 3:  $L_{\text{max}} \leftarrow \max_{t \in \mathcal{T}_{\text{top}}} \text{LEARNINGCOUNT}(t)$
- 4:  $N_{\text{pick}} \leftarrow \min(\lceil \alpha \cdot L_{\text{max}} \rceil, p)$
- 5:  $\mathcal{L}_{\text{pool}} \leftarrow \emptyset$
- 6: **for each**  $(t, s_t) \in \mathcal{T}_{\text{top}}$  **do**
- 7:   **for each** learning  $\ell \in \text{GETLEARNINGS}(t)$  **do**
- 8:      $\text{score}(\ell) \leftarrow s_t \times W[\ell.\text{valid\_level}]$
- 9:      $\mathcal{L}_{\text{pool}} \leftarrow \mathcal{L}_{\text{pool}} \cup \{(\ell, \text{score}(\ell))\}$
- 10:   **end for**
- 11: **end for**
- 12: Sort  $\mathcal{L}_{\text{pool}}$  by score descending; deduplicate by learning text
- 13:  $C \leftarrow \text{first } N_{\text{pick}} \text{ elements of } \mathcal{L}_{\text{pool}}$
- 14: **return**  $C$

---

### B Knowledge Base Schema

Each entry in the knowledge base is a structured record containing the following fields:

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**Algorithm 2** Hybrid Tool Retrieval 700

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**Require:** Issue description  $I_q$ ; knowledge base  $\mathcal{K}$ ; result count  $k=3$ ; dense weight  $\alpha=0.7$ ; sparse weight  $\beta=0.3$ ; validation weights  $W$

**Ensure:** Top- $k$  issue-learning pairs

```
1:  $q' \leftarrow \text{CLEAN}(I_q)$  {remove digits, normalize}
2:  $\mathbf{v}_q \leftarrow \text{EMBED}(q')$ 
3:  $\mathbf{s}_{\text{cos}} \leftarrow \text{COSINESIM}(\mathbf{v}_q, \mathcal{K}.\text{issue\_embeddings})$ 

4:  $\mathbf{s}_{\text{bm25}} \leftarrow \text{BM25}(q', \mathcal{K}.\text{index})$ 
5:  $\mathcal{C} \leftarrow \text{Top}(\mathbf{s}_{\text{cos}}, 100) \cup \text{Top}(\mathbf{s}_{\text{bm25}}, 100)$  {candidate pool}
6: for each  $i \in \mathcal{C}$  do
7:    $c_i \leftarrow \text{clamp}(\mathbf{s}_{\text{cos}}[i], 0, 1)$ 
8:    $b_i \leftarrow \text{MINMAXNORM}(\mathbf{s}_{\text{bm25}}[i], \mathcal{C})$ 
9:    $S_{\text{hybrid}} \leftarrow \alpha \cdot c_i + \beta \cdot b_i$ 
10:   $\text{score}(i) \leftarrow S_{\text{hybrid}} \times W[\mathcal{K}[i].\text{valid\_level}]$ 
11: end for
12: Sort  $\mathcal{C}$  by score descending
13: return first  $k$  elements of  $\mathcal{C}$ 
```

---

• **Task metadata:** task description, object type, action verbs, goal phase (SEARCH, ACQUIRE, TRANSFORM, PLACE, or RECOVER).

• **Issue-learning pair:** `issue_text` (the specific mistake encountered) and `learning_text` (the successful correction).

• **Validation level:** a confidence label reflecting downstream outcome:

- **VALID\_NEXT\_TRIAL:** the learning was applied and led to success in a subsequent trial (highest confidence). Weight  $\omega_{\text{next}} = 1.0$ .
- **VALID\_SAME\_TRIAL:** the learning is associated with success within the same trial. Weight  $\omega_{\text{same}} = 0.9$ .
- **CANDIDATE:** the learning was generated but the task was not completed (lowest confidence). Weight  $\omega_{\text{cand}} = 0.6$ .

• **Provenance:** references to the source task, trial number, and step range for both the issue and its resolution.

## C Experimental Details

### C.1 Dataset Statistics

**ALFWorld.** The full ALFWorld dataset (Shridhar et al., 2021) comprises 3,827 games split into

a training set (3,553 games), a validation-seen set (140 games), and a validation-unseen set (134 games). For knowledge base construction, we use *alfworld-mini*, a curated subset of 120 training games (20 per task type  $\times$  6 task types). We run Reflexion on this subset and extract approximately 400 issue-learning entries from the resulting trajectories using the pipeline described in Section 3.3.1. The validation splits are identical to those of the full ALFWorld dataset, ensuring direct comparability with prior work. No training games appear in the validation sets.

**WebShop.** We partition the WebShop environment (Yao et al., 2022) into a development split (200 tasks) and a held-out test split (200 tasks). The knowledge base is extracted from Reflexion training trajectories on a separate set of training tasks with no overlap with either evaluation split, yielding approximately 677 issue-learning entries.

**SQL diagnostic gate.** We include a paired SQL diagnostic gate of 50 tasks run with GPT-5.4 to probe transfer behavior in a formally regular action space. This gate is used as diagnostic evidence rather than as a full third benchmark. The paired run contains zero retrieved-help calls across memory variants and no environment-error matches in world logs, isolating memory and prompt effects from runner instability. Summary outputs are stored in the SQL transfer-radius experiment report directory.

### C.2 Framework Variants

We evaluate six framework variants in each environment:

1. **ReAct:** the base agent with no memory (Yao et al., 2023), serving as the primary baseline.
2. **ReAct + Reflexion:** the iterative self-correction baseline (Shinn et al., 2023), allowed up to 7 trials per task.
3. **ReAct + CR:** base agent augmented with Context Retrieval only.
4. **ReAct + TR:** base agent augmented with Tool Retrieval only.
5. **ReAct + CR + TR:** base agent augmented with both retrieval mechanisms.
6. **ReAct + hard-neg CR + TR:** ablation control that retrieves maximally *irrelevant* entries via inverted retrieval (Section 3.3.4).

All memory-augmented variants (CR, TR, CR+TR) execute a single trial per task, whereas Reflexion may execute up to 7 trials.

### C.3 Evaluation Metrics

**Task success.** The fraction of tasks completed successfully, computed as a binary outcome per task. For WebShop, this is corrected strict success.

**Average Steps per Trial.** The mean number of thought–action–observation steps per individual trial attempt. For Reflexion, this reflects the average length of a single trial, irrespective of retry count.

**Average Steps per Task.** The mean total steps per task across all trials. For single-trial methods, this equals Steps/Trial. For Reflexion, this accumulates steps across all retry trials (up to 7), capturing the full computational cost.

**Gain from ReAct.** The percentage-point difference between a variant’s task success and the ReAct baseline task success on the same split.

**SQL reward and regressions.** For the SQL diagnostic, we report exact task success, average environment reward, and paired regressions or improvements relative to ReAct on the same 50 tasks.

### C.4 Hyperparameters

Table 7 lists all hyperparameters used across experiments.

### C.5 Implementation Details

All agent reasoning uses Gemini 2.5 Flash routed through the OpenRouter API. We use greedy decoding (temperature 0.0) with a maximum output length of 2,048 tokens. API calls employ exponential backoff (2 s–30 s) with up to 8 retry attempts.

Retrieval embeddings use `gemini-embedding-001` via the Google GenAI SDK directly, with a batch size of 100 texts per API call. Embeddings are cached to disk alongside the knowledge base file; the cache includes a hash of the source file and the embedding model name for automatic invalidation on changes.

Each task trial is limited to 49 thought–action–observation steps across all agent variants. For Reflexion, this limit applies per trial, with up to 7 trials per task.

| Parameter                  | Value   | Description                        |
|----------------------------|---------|------------------------------------|
| <i>Context Retrieval</i>   |         |                                    |
| $p$                        | [5, 25] | Tunable cap on retrieved learnings |
| $\alpha$                   | 1.5     | Dynamic context sizing factor      |
| $k_{CR}$                   | 5       | Neighbor tasks retrieved           |
| <i>Tool Retrieval</i>      |         |                                    |
| $k_{TR}$                   | 3       | Results returned per query         |
| Dense weight               | 0.7     | Cosine similarity fusion weight    |
| Sparse weight              | 0.3     | BM25 fusion weight                 |
| Candidate pool             | 100+100 | Top-100 from each strategy         |
| <i>Validation Weights</i>  |         |                                    |
| $\omega_{next}$            | 1.0     | VALID_NEXT_TRIAL                   |
| $\omega_{same}$            | 0.9     | VALID_SAME_TRIAL                   |
| $\omega_{cand}$            | 0.6     | CANDIDATE                          |
| <i>Agent Configuration</i> |         |                                    |
| Max steps/task             | 49      | Episode length limit               |
| Temperature                | 0.0     | Deterministic decoding             |
| Max output tokens          | 2,048   | LLM generation limit               |
| Max Reflexion trials       | 7       | Retry budget                       |

Table 7: Hyperparameters used across all experiments.